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Disaster Preparedness and Response Education System for Schools and Colleges

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***Abstract*—Universities and colleges use emergency management practices that contribute significantly to safeguarding the safety, health, and welfare of students, faculty, and staff. The campus may encounter several types of emergencies including pandemics (e.g., COVID-19), natural disasters, and safety incidents. This document provides an overview of the types of emergencies that occur on educational campuses and how colleges and universities respond to the various types of emergencies. This work will also address recent advancements in technology being applied to disaster management through simulation and modeling; Internet of Things (IoT); communication and connectivity; and artificial intelligence analytics. Although these types of technologies provide exceptional value, they are not typically found in the real world or on a campus due to the lack of a common standard for integrating these technologies into an educational setting. Establishing a consolidated disaster preparedness and response education system that integrates interactive learning modules, simulations, real-time alerts, and performance measures will improve both the preparedness of students as well as institutional stakeholders, and their overall readiness, responsiveness, awareness, coordination, and resiliency.**

***Index Terms*—Emergency Management, Disaster Preparedness, Simulation, Education System, Public Health**

I. Introduction

In recent years, all types of educational institutions have become aware of the importance of preparing students with the knowledge and skills necessary to respond to disasters. Due to the potential vulnerabilities of school campuses to various types of disasters, ranging from natural disasters to fire emergencies and public health emergencies (such as the COVID-19 pandemic), the consequences can compound rapidly, if the institutions do not provide necessary disaster education that prepares students.

Emergency management responses of most educational institutions have been shallow, focusing mostly on administrative response rather than preparing students to be able to respond to disasters. As such, students have little knowledge regarding safety procedures, action plans during an emergency, evacuation procedures, etc., which exemplifies the need to develop a platform that educates and actively engages students in obtaining the knowledge and skill necessary to respond to disasters.

This project is developing a web-based tool designed to improve disaster education and also provide assistance in responding to disaster incidents for students. This web-based tool will use structured learning modules to educate students about how to handle various types of disasters. Each module will provide instructional content through video and allow for continued student engagement through quizzes and completion certificates.

To allow for proper assessment of training programs, the administrator dashboard will allow administrators to monitor student progress on the web-based tool. The administrator will have the ability to track the number of modules completed per student, the level of performance on quizzes, and the overall level of participation by each student. The administrator will use this information to assess the effectiveness of the training program and determine if students have been properly prepared for emergency situations.

By providing interactive learning, assessing performance and providing a monitoring and reporting system in one application, the proposed web-based tool will improve disaster awareness, preparedness and response capabilities of students. This level of preparedness provides improved individual safety,

as well as contributing to building a more resilient campus community.

II. LITERATURE REVIEW

The amount of bibliographic research related to the education of disaster management indicates that simulation technologies, Internet of things (IoT) systems, communication networks and artificial intelligence-based analysis systems (AI) will be critical tools in facilitating greater disaster preparedness and response efforts. These existing studies have been valuable resources that will enhance the development of a Disaster Preparedness Education System in both school and college settings.

The Dynamic Simulation of the Probable Propagation of a Disaster Within An Engineering System Using a Hybrid Network Simulation Model. [1]. Presented in IEEE, outlines a combined hybrid network model and Monte Carlo simulation to determine how disasters behave in complex systems. Vulnerable parts/components (interdependencies) can be identified and assistance can be provided to assist in making emergency management decisions; however, this study is restricted to an engineering case study and has limited extrapolation to other real life cases.

One approach to communication in the aftermath of a disaster makes use of Unmanned Aerial Vehicles (UAVs) for semantic communication in a way that reduces energy usage and provides continued connectivity, but the method primarily relies on simulation, so it has limited effectiveness when addressing security concerns Ideas in the Air: Unmanned Aerial Semantic Communication for Post-Disaster Scenarios [2].

Similarly, while a variety of technologies, such as 5G/6G networks, UAVs, and satellite networks have been analyzed for emergency communication, the lack of empirical data limits their applicability in a genuine disaster response.A Review of Applicable Technologies, Routing Protocols, Requirements, and Architecture for Disaster Area Networks [3].

The integration of the Internet of Things (IoT) into disaster mitigation as outlined by Yildirim .Design for Optimized Public Safety and Earthquake Disaster Mitigation with IoT [4], provides a real-time link between early warning systems and ShakeMap data to automate the identification of high-risk areas and trigger automated emergency response plans. However, despite positive pilot results, implementing such systems on a large scale may still present challenges.

The effects of the education of students on their evacuation behavior have been researched in the context of sustainability Considering the Effects of Sustainable Safety Education on Student Evacuation Behavior During College Emergencies [5]. The study found that mobile guidance systems and structured safety education greatly increased the stability of student recall and reduced evacuation time. However, the study had a limited timeframe and was conducted in a controlled environment.

The use of social media analytics as a means of increasing disaster awareness has been studied extensively, with a series of works including Risk Communication in Social Media During Emergencies: Information Switching Patterns [6], A Multistage Analysis of Risks in Social Media and Public Resilience of Impactful Crisis Communication During Flooding Emergencies [7], and The Evolution of Community Networks for Smartphone-Based Situational Awareness [8].These studies relate to how misinformation spreads, resilient response to a disaster, and how subjects change in times of disaster. Although these studies are useful for developing situational awareness, the study Text Quality Evaluation of Emergency Response Plans [9] utilizes Natural Language Processing (NLP) to improve clarity and consistency by applying Bi-LSTM-CRF to documents that contain emergency plans. Although the results of this study will allow emergency response plans to be written more clearly and consistently, they rely on training data of good quality and specific to the language. In the study Dual Hackathon Based on Immersive Simulation [10], a significant difference was found in students' ability to both manage technical challenges and develop risk management skills through immersive simulation-based learning but is completely dependent on institutional support.

The reviewed literature on the disaster simulations, Internet of Things (IoT) monitoring, emergency communication, social media analytics, and safety education fails to identify a digital learning platform based on a curriculum that integrates all components of the disaster response system. Therefore, the proposed project will develop an integrated educational platform following several integrated learning modules, virtual drills, real-time regional alerts based on geographic location, and preparedness tracking into a single educational platform for all students enrolled in an accredited institution of higher education throughout India.

III. METHODOLOGY

Experimental Design

1) ***Pre-Experiment Preparation****:* To assess the effectiveness of a web-based learning platform on improving students' disaster preparedness, an experiment was set up. All student participants were briefed on the purpose of the study and instructed on how to navigate the web-based learning platform (including links to videos and completing quizzes). Participation in the study was voluntary. Students were also told that they could withdraw from participation in the study at any time, as seen in the Fig.1.

2) ***Implementation of the Educational Intervention****:* Students assigned to the experimental group were given access to a web-based learning platform, which contained instructional modules consisting of video-based content centered around various types of disaster scenarios and safety measures. The interactive design of the web-based learning resource allowed for enhanced understanding for students using visual supports and assessments combined.

The intervention included a prescription-gated walk-through of instructional modules with each student progressing through the modules in a systematic fashion. The use of trainings with an emphasis on interactive components such as quizzes, simulation scenarios, and case studies helped to maintain

Fig. 1. Fire-fighting equipment in University/Colleges

Fig. 2. Evacuation safety knowledge education

student engagement. In addition, the students received an opportunity to actively participate in virtual simulations and apply their knowledge from the training modules in actual simulations. Students also received immediate performance feedback at the conclusion of each activity, providing them with information regarding their strengths and weaknesses.

*3) **Evaluation and Assessment**:* Students started out by doing a pre-assessment quiz before beginning their lessons to measure what they knew beforehand; upon completion of the lesson material they were given a corresponding post-assessment measure to evaluate their understanding of the subject matter that was taught to them as well as whether or not they retained any of that information.

Students performance prior to and following this intervention was analyzed to assess if the website-based learning methodology was effective in improving disaster awareness and preparedness.

The evaluation process also included a comparative analysis of pre-test and post-test scores to assess improvements in knowledge in addition to statistical analyses that identified significant differences in learning outcomes. Various user engagement metrics including the length of time that was spent on each module of study, quiz completion rates and participation rates in virtual drills were tracked and analyzed during the evaluation. Moreover, each student evaluated his/her overall experience with respect to the usability of the website; clarity of the material that was presented to him/her; and his/her overall learning experience. Each of these measures collectively yielded a comprehensive evaluation of how effective the experimental system was; as represented in Fig. 2.

IV. SYSTEM DESIGN AND IMPLEMENTATION

A. System Architecture

*1) **A. Conceptual Block Diagram of the System Architecture**:*

- **Users (Students and Teachers)** The students and teachers access the program to learn about what to do in the event of a disaster as well as participate in drills and take quizzes based on their knowledge of the subject.
- **Login and Registration Module** This module allows the user to create an account and log into the system with their login credentials. The User Authentication System will handle all user authentication and store all user profile data including Name, Role (Administrator or End-User), Institution (School or State), and Region (Rural or Urban).
- **Dashboard Module** The Dashboard will be the main user interface to view user progress, view modules that were completed, view drills that were completed, and view user's preparation level.
- **Learning Modules** Educational modules for several natural disasters (e.g. floods, earthquakes, cyclones, landslides) are available through this module. Each module contains safety issues, preventative measures, and educational materials about preparing for each natural disaster.
- **Virtual Drills** The Virtual Drill Module simulates different emergency situations such as fire drills/ evacuation, earthquake drills/ evacuation, and flood drills/ evacuation. The purpose of this module is to provide users with simulation training to follow the correct procedures for responding to a disaster.
- **Emergency Contacts Module** The Quick Reference Module provides immediate access to some of the most important National Emergency Telephone Numbers. The emergency numbers include: 112 (National Emergency), 100 (Police), and 101 (Fire Services).
- **Quiz and Assessment Module** The Creation of a quiz will be required for preparation modules and for drilling to determine how much each participant learned about disaster preparedness principles.
- **Admin Panel** The Administrator Panel will allow Administrators to monitor student activity, and will enable the Administrator to identify students' previous history related to quiz results, and assess their preparedness levels.

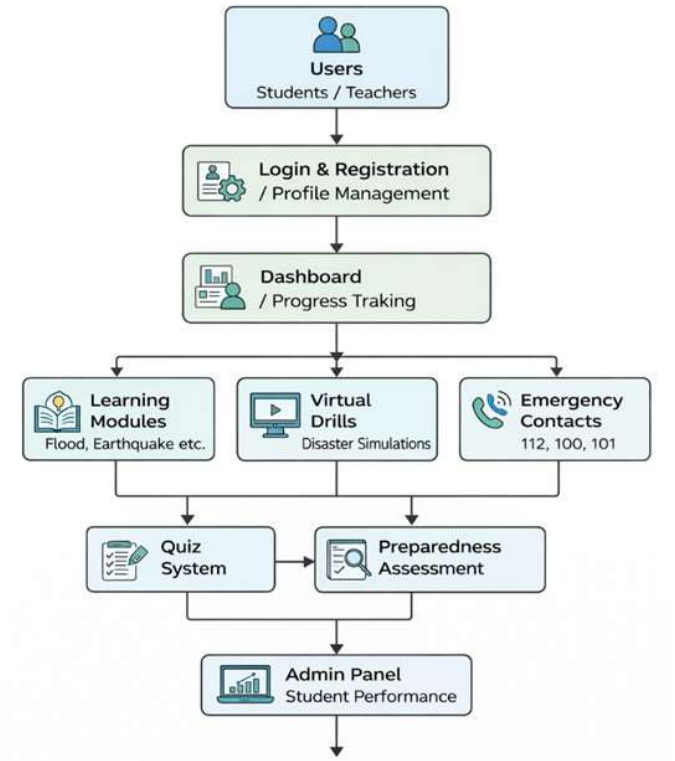

Fig. 3. Block Diagram of the System Architecture

B. *System Architecture and Data Flow*

In summary, this architecture integrates learning, training, assessment and monitoring in order to improve disaster preparedness for the educational community, including both students and staff.

- The data flow for the Disaster Ready India system will be structured, which will allow efficient communication between the user interface (UI), application logic (AL) and database. This structured data flow will allow the Disaster Ready India system to provide efficient delivery of disaster preparedness information and track user activity.
- The user will log into the system or select a specific feature (learning module or virtual drill), which will initiate a request to the back-end server through the browser. After receiving a request from users (i.e., front-end clients), back-end servers typically do not send back a completed document. Instead, back-end servers first determine which types of required data will be used to deliver the requested documents. If the data exists in the database, it will be returned to the user from the necessary tables (i.e., user profiles, quiz scores, disaster modules, preparedness records).
- If flexible content exists (i.e., disaster descriptions, safety information, multimedia content) in structured files, the back-end server will load the required data and prepare it for display. This allows the Disaster Ready India system to handle a variety of different types of educational content, while also enabling future updates and revisions
- After collecting the requisite data, the Backend communicates back to the Frontend interface with its response so that the Frontend can organize and display the information in a clear, user-friendly manner using a dashboard, learning modules, drill simulations, and quizzes, enabling Users to receive accurate information quickly, and participate in activities without delays.
- In order to enhance overall system performance, lightweight optimization techniques have been applied to reduce loading time and enhance system user experience by making it more responsive. The materials contained within the Learning Modules that Users access frequently as well as Learning Module data have been designed for efficient processing, allowing the system to work smoothly with multiple Users accessing it at the same time.
- In general, the structured data flow of the system's design aids in maintaining system reliability, enhancing User Experience, and providing an efficient means of delivering disaster awareness and preparedness training Content.

C. *Implementation Details*

The system was implemented using an incremental development process to achieve reliable and efficient results. The following steps were utilized in the implementation of the system:

- Backend System was designed and developed ii a way to include a structured set of database tables that will contain the needed information about Users, quizzes, and the various types of disaster-related Content.
- Frontend User Interface was designed and developed to be user-friendly (ex. Login page, dashboard, Learning Modules, drill simulations, and quizzes).
- Integrated frontend and backend to enable real-time data interaction and updates.
- Incorporation of multimedia (images and video) increased the educational value and engagement of Users.
- Testing and debugging were completed to verify and correct flaws.
- Optimized performance to increase the system's responsiveness as well as reduce its load time.

The entire process has provided a more scalable, interactive and efficient method of educating individuals on how to prepare for a disaster.

V. ALGORITHMS AND COMPUTATIONAL METHODS

The Disaster Preparedness and Response Education System (DPRES) is a fully online system that can be accessed via the Internet by schools and colleges. Therefore, no

offline hardware devices are needed or physical sensors to be installed. All aspects of the students' education i.e., learning, submitting assignments, tracking their performance, communicating with others during an emergency, and all administrative functions relating to monitoring students, can be performed using an Internet-based system that is centrally located.

Students can access the platform via their computers or mobile phones using a web browser. Because it is based in the Cloud, all data is stored and processed on Cloud servers giving all users real-time and remote access to the data.

*1) **Online User Authentication System**:*

- Students and administrators will have secure log-in capabilities.
- Students will log in to find information on disaster preparedness and submit assignments access disaster awareness materials and submit assignments**.**
- Administrators will log in to monitor the academic progress of their students and send alerts.
- All authentication processes will be completed through online database verification.

*2) **Web-Based Learning and Training Modules**:*

- Disaster education content includes:
  - Fire safety
  - Earthquake response
  - Flood precautions
  - Emergency evacuation steps
- This information will be delivered using fundamental online modules that include written resources and visual resources (e.g. Videos).
- Students will be able to access the resources at any time with an available Internet connection.

*3) **Online Assignment and Quiz Evaluation**:* The evaluation of objective-type quizzes is by the automated system. Results are stored in a secure online database. After submitting their objective-type quizzes, students can quickly see their results; thus, enabling them to have a better comprehension of disaster preparedness concepts.

*4) **Online Emergency Alert System**:* Administrators can send an emergency alert through an online dashboard interface. When triggered, an emergency alert is sent to students logged in through the online dashboard interface.

*5) **Real-Time Location Feature (Online-Based)**:*

- If permission to access location services is granted, the system can determine the locations of users using web browser-based services.
- Assists in determining whether students are on campus during emergencies.
- This function is only operational when there is some form of internet connection available to the user.

VI. RESULT AND DISCUSSION

The proposed system, "Disaster Ready India," was developed as a digital platform integrated with various modules to increase the capacity for disaster preparedness and management in educational institutions, including both schools and colleges.

The integrated modules consist of multiple elements that cover disaster awareness, disaster response, and disaster preparedness information provided in an organized manner on a digital platform. So, the system is set up using one of the following five major modules: the dashboard, learning modules, virtual drills, emergency contact information, and end-user account creation and profile management.

By using technology-connected learning tools, the system has demonstrated how it can enhance the ability for students and staff to prepare for disasters. All of the modules were tested for usability, functionally and ultimately effectiveness in promoting disaster awareness and safety through education.

A. System Implementation Results

*1) **User Registration and Profile Configuration**:* The system will allow all users (students and teachers) who register by entering primary details such as name, user role, school name, and area of residence to receive a recommended learning content which focuses on the region's specific disaster risk. For example, a school located in a coastal region would receive a considerable amount of content associated with preparing for a cyclone.

The results indicate that region-based customization significantly improves the relevance and effectiveness of disaster education content. The structured registration process also ensures proper user segmentation and enables the system to maintain personalized preparedness records for each user.

*2) **Dashboard Analytics with Location-Based Monitoring** :* The Student Dashboard Module offers a custom interface that allows users to follow their progress, level of interaction, and level of readiness within the system. Users will see key indicators of their performance displayed on their dashboard including number of modules completed, quiz scores, number of drills participated in, and total academic performance. The information displayed on the dashboard updates automatically as the time student interacts with the system. Users will begin at the level of beginner but will advance to advanced levels as they complete additional educational content and practice on a regular basis.

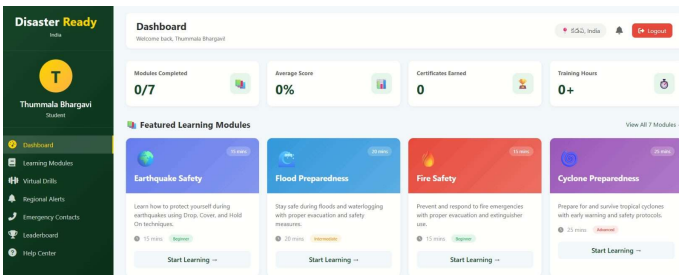

Fig. 4. Enter Caption

Another unique functional characteristic of users is that the dashboard collects geo-based information about each user and

provides content that is based on the user's geo-based location. This way, users have access to an array of specific content regarding disasters based on their geo-based location (e.g., users in areas prone to earthquakes, vs. users in flood-prone areas).

Using such an approach allows the student to learn relevant, practical content; continuous monitoring of the user's activity aids in improving student engagement, retention of knowledge, and overall preparedness for future disasters as shown in Fig.4 .

*3) **Learning Modules**:* The Learning Module section provides users with well-structured instructional materials about various types of disasters. Some examples of modules are: Earthquake Preparedness, Flood Preparedness, Fire Safety, Cyclone Preparedness, and Landslide Preparedness.

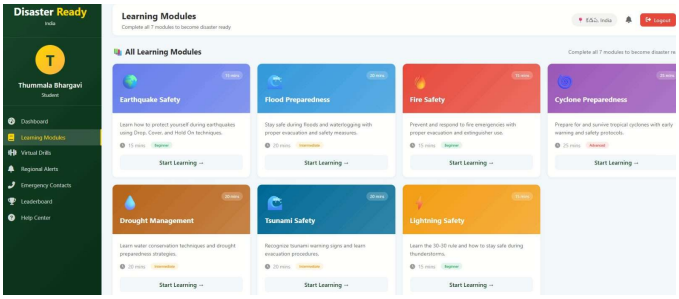

Fig. 5. Learning Modules

Each module is categorized according to the level of difficulty (beginner, intermediate or advanced) and takes approximately 15 to 25 minutes to complete. Users acquire knowledge slowly and gradually through a modular method known as micro-learning because they are not inundated with too much information.

Experimental results have demonstrated that modular learning promotes increased levels of engagement and retention of knowledge, due to the inclusion of Recommended Tags that delineate hazards unique to the regions as shown in Fig.5.

*4) **Virtual Drills**:* Through simulation-based drills, users can participate in simulated "real-life" drills in emergency situations, such as; fire evacuation, flood evacuation individuals will receive training on how to respond appropriately and quickly during emergency situations.

Results from testing have shown that by repeating these drills, participants develop greater skills when making decisions, and that their reaction times during simulated emergencies were reduced. The virtual drill platform allows the linking of both theoretical education with practical disaster response training.

*5) **Emergency Contacts Module**:* Users will have easy access to National Helpline Telephone Numbers, when required in the event of emergencies, through the Emergency Contacts Module. The module will offer an easy way for users to access 3 of the most common emergency service

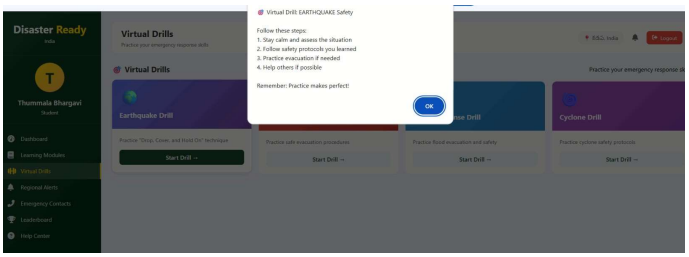

Fig. 6. Virtual Drills

numbers, which include the National Emergency Number 112, Police Services 100 and Fire Services 101.

By integrating Emergency Contacts/Numbers directly into the platform within the Emergency Contacts Module, users will have quick access to essential services. Consequently, the usability of the system will be enhanced in terms of real-life application, and institutions' level of preparedness in managing hazards will be increased.

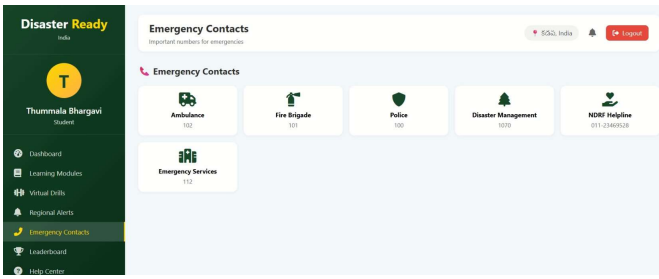

Fig. 7. Emergency Contacts Module

*6) **Quiz-Based Learning System**:* Through the interactive quiz, "The Earthquake Safety Quiz," students take a timed assessment consisting of ten questions to meet the required 70 minimum score to pass (or complete).

*7) **Certificate Generation**:* The Certification of Completion will automatically be generated for students who complete the Quiz Assessment.

Features:

Student name and email Training module (Earthquake Safety) Score and percentage Issue date and certificate ID

Observed result:

Certificate issued with 90 score and PASS status

This ensures:

Recognition of learning achievement Encourages student participation Adds value for academic and professional use as shown in Fig.9

B. ADMINISTRATOR MODULE RESULTS

The Administrator Module is the central core of all Admin features which provides real-time data and Insight into how

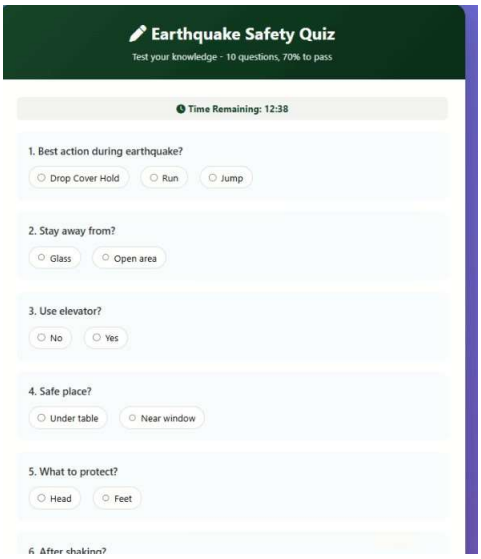

Fig. 8. Quiz-Based Learning System

Fig. 9. Certificate Generation

well the system is performing, what users have Access to, and how well they are achieving the learning outcome goals of the system.

1) ***Dashboard Monitoring and Analytics****:* The implementation of the admin dashboard has proven to be successful because it is very efficient, provides data-driven results, is easy to understand, and is user-friendly.

Features of the admin dashboard allow real-time updates of several summary items for example:

- All of total number of attempts to complete a quiz - Current number of students (active) using the system - Average scores of all students - Results of all completed quizzes

The information observed from the administration dashboard include:

- The system correctly tracked participation of students (1 student). - The average score for all students that completed the quiz was 90 which demonstrates a successful outcome for students' learning. - Earthquake module had statistics of participatory activities for Earthquake safety, while all remaining modules could be completed at a future date.

Therefore, the dashboard appropriately portrays participation rates and preparedness for the disaster of the Earthquake.

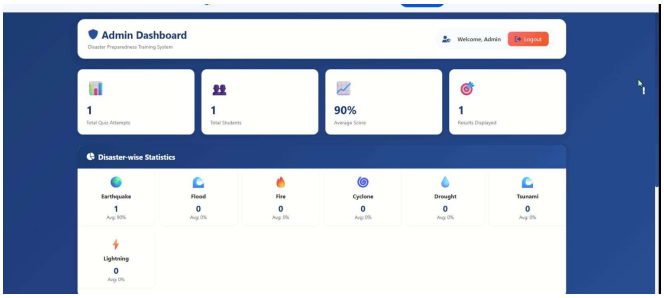

Fig. 10. Dashboard Monitoring and Analytics

2) ***Disaster-wise Performance Analysis****:* Real-time metrics like student engagement, system performance and output come together to form the Admin Dashboard as the primary reporting tool of the system. The Admin Dashboard reports on key metrics including: total number of quiz attempts, active student count, average score as well as completion percent of all attempted quizzes; these are all reported in an easy to read, user friendly format.

The usefulness of this tool to track and report on student participation was demonstrated via one active student with an average score of 90 indicating a successful level of output from the student's learning experience. Additionally, the Admin Dashboard reports out to show student engagement by Disaster type including: Earthquake, Flood, Fire, Cyclone, Drought, Tsunami, and Lightning; thereby providing administrators the opportunity to focus on low engaged areas and raise awareness regarding content created for these types of Disasters and improve upon them.

3) ***Emergency Alert System****:* The Send Emergency Alert tool enables administrators to send real-time safety messages, which will appear immediately on the student dashboard. The first example could be "Rainfall warning today!" and is shown to the administrator as a preview prior to being sent out, so that they can determine if everything looks correct. After they have published this alert, it will be displayed prominently on the student interface so that users will know that there is currently a risk.

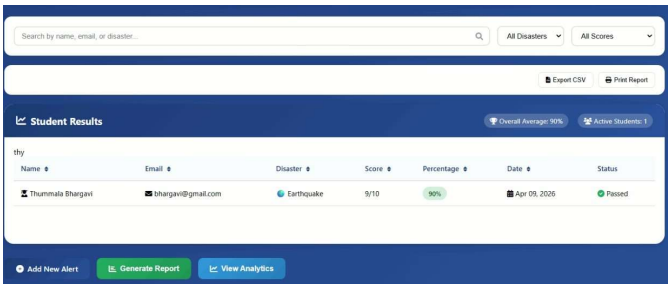

Fig. 11. Disaster-wise Performance Analysis

The intent behind these alerts is to be clear, concise, and easily visible so that students are able to understand their situation rapidly and make an appropriate decision. Because alerts are integrated into the dashboard, students do not have to worry about missing out on important information while they work in a learning module or take a quiz.

Results show that this tool improves fast, reliable communication in emergencies; decreases the time for responses to emergencies; and increases the overall preparedness of students for real-time disasters by keeping students up to date on the status of active disasters.

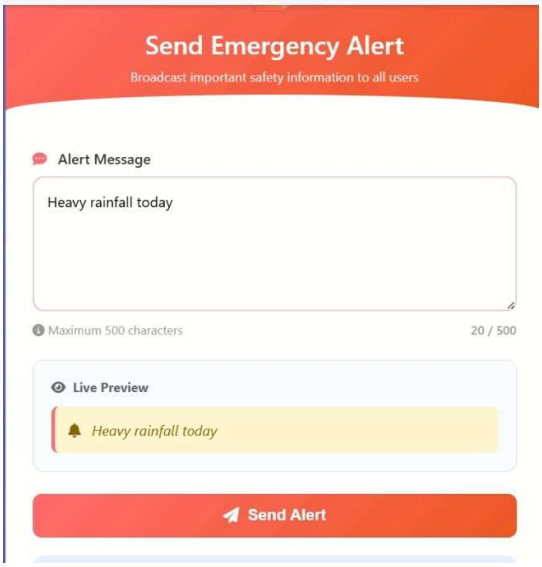

Fig. 12. Enter Caption

4) ***Report Generation****:* The Generate Report feature, administrators can create and generate reports about data for analysis in structured formats, like exportable files (CSV). The data in the reports will include information on the total performance of all users, provide access to key metrics such as student performance, quiz performance, and trends in participation, and allow administrators to track the progress of users over time, identify high-performing and low-performing users, and measure the effectiveness of all modules. Reports are also visually appealing, making them easier for administrators to read, which provides documentation and improved transparency for educational institutions, enabling them to make more well-informed decisions about improving training and disaster preparedness. Reports can also be shared with stakeholders for the purpose of review and planning, and used to compare the performance of different types of users across different periods of time. The overall outcome of this feature is that it strengthens the monitoring capabilities of the system as well as its ability to continuously improve..

Fig. 13. Enter Caption

5) ***View Analytics****:* The Analytics View Module enables users to see greater detail than just the averages of total performance, number of active users, and trends related to learning progressions these help facilitate deeper analytical understanding. These analytics give a complete view of how users engaged with the learning material, how the user learned, how they behaved while learning, and what was being taught. From these results, administrators can see where their users are excelling and where they need improvement.

The use of data is helping institutions to make better decisions about what to update, how to train their users, and how to help their users engage. The View Analytics Module also helps measure the effectiveness of the training by measuring how much the users' performance and preparedness levels improved to provide users with more opportunities to better handle real-life disaster situations after they complete the training.

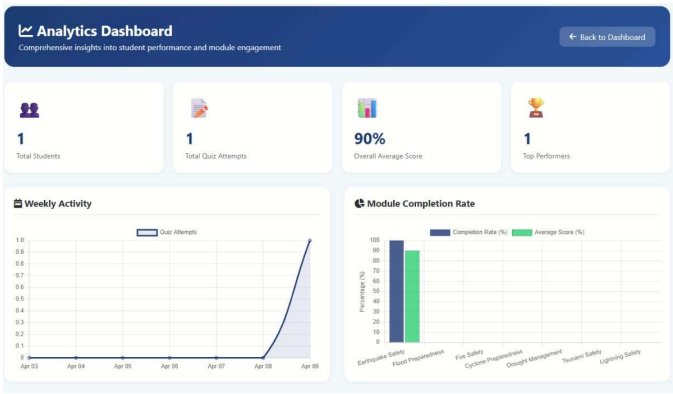

Fig. 14. Analytics Dashboard

C. *DISCUSSION*

The findings show that the inclusion of education, simulation and emergency communication into one digital platform maximizes the capacity of educational institutions to be prepared for disasters.

1) ***Improved Awareness****:* The Structured Learning Modules build the student's and staff member's knowledge of hazards. The topic based classification provides for the full spectrum of disasters that occur regularly within India.

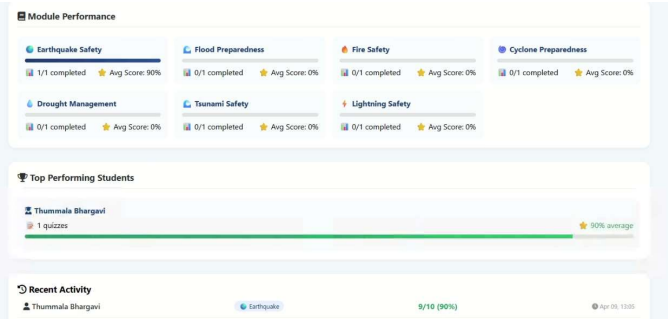

Fig. 15. Analytics Dashboard

2) ***Practical Preparedness****:* Virtual drills facilitate skill development and real-time emergency response practice within a simulated setting. Such drills enhance one's capacity to act without fear, developing their confidence in responding in real emergencies.

3) ***Regional Customization****:* Recommendations based on location ensure that emergency preparations are pertinent to the region's hazards (e.g., those preparing for cyclones do so differently). This type of learning is essential for India's varying and unique physical geography.

4) ***Performance Monitoring****:* The Dashboard Analytics System creates motivation for the users with measurable progress indicators. Preparedness Levels (beginner to advanced) create a 'gamified experience' for the learner.

5) ***Institutional Implementation Feasibility****:* The web-based Disaster Preparedness and Response Education System is designed to be modular, scalable, and economical for both schools and colleges. It has the capability to be part of both existing academic management systems with minimal infrastructure requirements.

VII. CONCLUSION

The modular structure of the system makes it easy to grow to accommodate all the needs of educational institutions. The systems that are already in place can also connect to or interface with the new system without requiring special hardware or software to run the complete system.

For students, the system provides a means of logging in securely, accessing disaster preparedness training materials, taking quizzes and submitting completed assignments. Once a student finishes a quiz, they will receive immediate feedback that can help them understand the level of learning they have accomplished, as well as how to continue building their understanding of how to respond to disasters.

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- Individual student quiz scores
- Assignment submission status
- Participation in disaster awareness modules
- Overall engagement levels
- Completion rates

The dashboards allow teachers to analyze student performance in terms of quiz completion, determine weak areas for the student and the overall amount of disaster preparedness training completed by all students. Collectively, the dashboards will decrease the amount of time spent on manual assessment by teachers and create transparency in the tracking of students academically.

The emergency alert feature will provide the ability for teachers to send out real-time alerts to all students and quickly communicate to them in the event an emergency occurs.

The overall results demonstrate that the system works effectively to supplement student learning and provide mechanisms for teachers to effectively monitor and assess their students all through a single online platform.

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